

Desigualdad de Bessel

Teorema 1 (Desigualdad de Bessel). Sea $(H, \langle \cdot, \cdot \rangle)$ un espacio prehilbert y $\{u_n\}_{n \in \mathbb{N}}$ un sistema ortonormal en $H \implies \forall x \in H : \sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2 \leq \|x\|^2$.

Demostración: Calculamos la norma del vector $x - \sum_{n=1}^N \langle x, u_n \rangle u_n$:

$$\begin{aligned}
 0 &\leq \left\| x - \sum_{n=1}^N \langle x, u_n \rangle u_n \right\|^2 = \left\langle x - \sum_{n=1}^N \langle x, u_n \rangle u_n, x - \sum_{m=1}^N \langle x, u_m \rangle u_m \right\rangle \\
 &= \|x\|^2 - \sum_{n=1}^N \overline{\langle x, u_n \rangle} \langle x, u_n \rangle - \sum_{m=1}^N \langle x, u_m \rangle \langle u_m, x \rangle + \sum_{n=1}^N \sum_{m=1}^N \overline{\langle x, u_n \rangle} \langle x, u_m \rangle \langle u_n, u_m \rangle \\
 &= \|x\|^2 - 2 \sum_{n=1}^N |\langle x, u_n \rangle|^2 + \sum_{n=1}^N |\langle x, u_n \rangle|^2 = \|x\|^2 - \sum_{n=1}^N |\langle x, u_n \rangle|^2.
 \end{aligned}$$

Por tanto, $\forall N \in \mathbb{N} : \sum_{n=1}^N |\langle x, u_n \rangle|^2 \leq \|x\|^2$. Tomando límite cuando $N \rightarrow \infty$, se obtiene la desigualdad de Bessel. ■

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